

Environmental Data Science: Homework #4
Matrices, arrays, and working with spatial data
Due: Before Class on 9/23

Ozone Analysis

Using the same dataset we explored in class (NASA GEOS-CF):

- i. Plot a map for O₃ concentrations in parts per billion.
- ii. Answer the following questions:
 - a. Where is O₃ qualitatively high?
 - b. What latitude and longitude is O₃ highest?
 - c. Where is O₃ qualitatively low?
 - d. What latitude and longitude is O₃ lowest?
 - e. What is the O₃ concentration over LA?
 - f. What is the O₃ concentration over Hawaii?

Final Project Pitch

The goal here is to start thinking about what final project analysis you'll be doing. To that end, describe:

- i. What data you'll be working with. Where do you get the data? How much data do you have (rough #)
- ii. What type of analysis do you want to do? For example:
 - a. Regression
 - b. Classification
 - c. Emulation
 - d. Scientific Discovery
- iii. Why could this analysis be scientifically important?

Submit your code via GitHub. A PDF submitted to Brightspace is also appropriate if you don't want to pitch your final project in your jupyter notebook.